

A modern cloud based recycling system for smart cities

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Abstract—The rate of urban waste production is constantly increasing as a result of the Earth's rapid population growth and modern lifestyle. Organic and non-organic recyclable waste, which accounts for a significant portion of urban solid waste, has raised environmental concerns. The best way to create a healthy ecosystem is to recycle. According to estimates from the US Environmental Protection Agency, approximately 75% of total waste can be recycled, but we only recycle 25% of it. The majority of the non-recyclable waste is dumped on the ground or incinerated. We can save energy and create a valuable by-product that can be used as a locally generated fertilizer through recycling organic materials through the compost process, whereas non organic materials can be used as industrial materials. Under this study, we suggest a solution: a low-cost and efficient Smart Recycle Bin that aides the recycling process by using modern technologies such as environmental sensors and the LoRaWAN protocol. A centralized Information System gathers data from smart bins that can be placed virtually anywhere and can help with waste collection. The Information System will generate real-time routes for waste collection vehicles. We ran several tests with our smart bin prototype, assessed its performance, and came to the conclusion that it is a viable solution.

Index Terms—smart-bin, algorithm, LoRaWAN, recycle, compost, reusable materials

I. INTRODUCTION

Food wastage has become a major issue affecting humans as a result of the constant growth of the population and the lack of ability to exploit accessible natural resources. It represents significant nutritional, environmental and social issue. As the recycling data from the United States Environmental Protection Agency indicates, 75% of the total Municipal Solid Waste (MSW) generated and could be recycled is wasted or lost [1]. The problem of recycling affects all humans at a global scale and it is directly linked to the contamination of the environment, the growth of our economy and even social standards.

The process of recycling consists of two main categories: (1) the recycling of organic materials (such as food that would have otherwise been wasted) and (2) the recycling of non-organic materials, such as paper, plastic, aluminum and glass. Even though there is progress being made, regarding the traditional recycling of non-organic materials, the recycling or composting food waste, which accounts for 37.57% of the total MSW generated per year [2] is only recently starting to become popular.

The composting process, which can be explained as the process of biological stabilization and decomposition of organic waste material under specific aerobic conditions [3], can generate a final stable product out of organic material, with the main characteristic of non-phytotoxicity; free of pathogen species and suitable for use as a plant growth substrate and nutrient source, as well as a biofertilizer to enhance soil properties due to the ability to use compost as a fertilizer, composting is the most popular method of recovering content from the organic fraction of municipal solid waste. This approach tends to be the proper technique for solid waste management, since it efficiently eliminates organic waste. Composting is seen as a way forward in the Eu Countries to fix both nutrients conservation and organic matter availability, thus improving soil conditions.

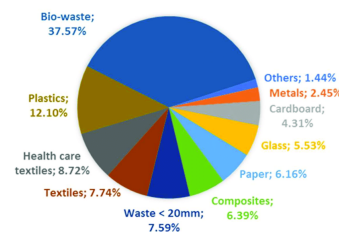


Fig. 1. Global Municipal Solid Waste generation composition for 2019

The management of MSW is directly correlated to the health of the environment. Excess amounts of waste materials are usually burned, buried or thrown away in landfills or even the sea. As a result, the air, soil and water are polluted and the flora and fauna are disappearing. This effect is particularly evident in developing countries, where the recycling rate is even lower. Research suggests that in some developing countries, the toxins and chemicals in the air have concentration many times above the acceptable rate. Living in an environment with so high concentration of toxins in the air, is only a matter of time until the population starts experiencing respiratory and neurological diseases.

Given that a large amount of municipal solid waste can be recycled, it is important to utilize recycling techniques in order to reuse materials in order to minimize environment problems. Many countries in the European Union are making progress regarding recycling, mainly by implementing laws to motivate citizens to recycle and make non-recycling harder. In most

cities, for example, traditional waste collection bins that in the past were used to collect all materials have been replaced by single-material bins. Legislation prohibits the disposal of different, other than the designed materials in these bins. As a result, even if these bins simplify the recycling process for the municipalities and the waste management authorities, it makes recycling harder for the civilians.

The composting process' effectiveness is determined by three key factors: heat, humidity, and oxygen concentration. Heat describes the behavior of microorganisms at various levels, humidity makes nutrients biocompatible, and oxygen dictates the aerobic state. To achieve nutrient-rich compost, these three criteria must be monitored during the composting process. This is not always simple, however, since it typically necessitates the use of expensive and large monitoring equipment. This method also emits a foul odor, which usually deters people from composting their food at home.

This research work extends our previous paper [4]. Moreover, we have developed a system that can utilize real-time data from the smart-bins and calculate paths for waste collection vehicles, in order to further enhance the efficiency friendliness to the environment. People in need of fertiliser can use the IS to locate the nearest smart bin. The smart-bins outlined here can be installed in prominent locations in the city to aid in the recycling process.

The remainder of the paper is organized as follows: The related research of this paper is summarized in Section II, along with related work. The proposed hardware and software implementation is presented in Section III. The experimental conclusions are discussed in Section IV. Finally, section V summarizes the study's findings.

II. RELATED WORK

There are several research papers on smart bins in the literature. The majority of them, however, are concerned with waste separation, waste classification, and waste management in general. Many researchers haven't looked into smart bins that promise to help with the composting process. As a result, the literature review isn't only about smart compost bins; it's also about smart bin implementations in general.

The authors of *et al.* [5] suggested a prototype which sent notifications when the fullness was 100. The authors utilized an intelligent sensor that could detect whether or not the bin was full. As the bin's lid closes, the sensor takes measurements. By comparing two consecutive measures, their program can also calculate the amount of waste obtained each time it is opened. Despite the high quality of the work presented, most of the reviewed works have important limitations, particularly on the software and practicality side.

The main difference between our work and that of the others is that we concentrate on promoting the composting process. The bins' embedded device is primarily utilized to capture and transmit information to the cloud server. As a result, the proposed system's computational requirements are smaller than those of other implementations, allowing us to use a low-cost embedded device in the bins and have production costs

as low as possible. This is one of many prototypes we created while looking for better ways to increase recycling rates.

III. IMPLEMENTATION



Fig. 2. The proposed Smart-Bin prototype

A Raspberry Pi Single Board Computer (SBC) embedded system and a LoRa platform are the key components of our smart bin project. The embedded device that we selected is suitable for our project, owing to its low cost, which is critical for the project's use cases. The Raspberry Pi Zero W has a single core ARM Central Processing Unit (CPU) with Wi-Fi and Bluetooth built in. Since our project's main processing unit is the server that hosts the IS, the Pi's processing resources are sufficient for our needs (Information System). Food waste may be deposited from the top into the bin (Fig. 2). The camera module captures an image every few minutes, and the mounted sensors (temperature and pressure) take measurements. The embedded device is responsible for compressing and encrypting the captured measurements before transmitting them to the server, located in the cloud.

Captured information is being deleted from the bins after the cloud-based IS receives them. If any unsafe situations are observed when the data is being processed by the IS (such as extreme temperatures and fire indications), the cloud system automatically activates the warning system that is installed on the bins, to prevent and disallow residents from using them. We have also developed an optional feature to automatically inform the relevant people, like the local officers to investigate the situation. An ultrasonic sensor is located near the entrance point of the Smart-Bin. This ultrasonic sensor is used to facilitate in the detection of the container's fullness percentage. Smart-Bins lock and prevent people from using the devices when the container's fullness percentage exceeds the maximum permitted level, using an electromagnetically operated seal.

A. Connectivity

The relation between Smart-Bins and the Information System is also a significant improvement over other implementations. We chose to use LoRa in this project. Semtech's LoRa is a patented spread spectrum modulation technique that can work with any MAC layer. LoRaWAN is a proposed MAC that uses a simple star topology to run a network. The LoRa system consists of the following components: (1) LoRa End-devices, (2) LoRa Gateways and (3) LoRa NetServer. The LoRa protocol has transmission speeds ranging from 0.3

kilobits per second to 11 kilobits per second. This is the best option for our project's requirements. It has a wider coverage range than WiFi. Mesh methods could not be used because of the large distance between two bins. The GSM network's operating costs make it an inefficient protocol for our needs. The recorded images and sensor measurements are temporarily saved to the Raspberry Pi Zero W's flash storage before being encrypted, compressed, and finally transmitted to the Information System through LoRa. Each measurement is collected by the IS and then deleted from the Raspberry Pi. Fig. 3.



Fig. 3. The LoRaWAN communication system being utilized by the bins; multiple bins are being connected to the main LoRAWAN gateway and transmit the captured temperature and pressure sensors along with images of the inside of the bin.

B. Software

One of the proposed smart-bin's notable aspect is its ability to be controlled remotely. The collected photos and measurements are sent to the Information System (IS) at a predetermined and user-configurable interval. The combination of the sensors we used, along with the camera module are necessary in order to constantly check the contents of the smart bin, and prevent hazardous situations. To prevent third parties from accessing transmitted data and reduce transmission time, it is critical that the data be compressed and encrypted before the link is formed. Lossless formats like WebP and PNG, as well as lossy formats like JPEG, can be used to compress images. We decided to use an asymmetric algorithm with public and private RSA keys to encrypt the transmitted data. [6], [7].

The Smart-Bin prototype that we build for this project is accompanied by the Information System, a web platform that allows local authorities and stakeholders to remotely manage the installed bins. For the development of the system, we mainly utilized software tools available in the public domain, like MySQL and Php. A specific serial number is assigned to each smart bin in the system. More technical knowledge about the Information System, such as algorithms and code diagrams, is outside the reach of this paper and will not be addressed.

C. Waste Collection Pathing

For a truly efficient and smart recycling solution that can be deployed in modern cities, besides the smart-bin devices

themselves and the technologies they are using, we also need a system that can calculate waste collection paths in real time, so that the vehicles that collect the contents of the bins have efficient routes. The traditional method of collecting the contents of waste bins, is at predefined intervals (for example every day, or 3 times a week) at predefined paths. This is inefficient, as some bins may be empty or may contain very small amounts of waste. As a result, valuable resources (such as time, work-hours and gas) are wasted. In this paper, we propose an algorithm, that using real-time data provided by the deployed smart-bins, we can calculate efficient paths for the waste collection vehicles.

At first, the algorithm selects the locations of all deployed smart-bin devices, except those that are empty or are less than 50% full, and using the maps API (Application Programming Interface) from Google, it calculates the driving distance of each bin to every other bin. These calculated distances (Eq. 1) are then added to a multi-node graph.

$$D_{total} = \{D_1 + D_2 + \dots + D_i\} \quad (1)$$

Feeding Dijkstra's [8] algorithm with the multi-node graph, we can calculate the most efficient route for the waste collection vehicle in order to traverse from one deployed smart-bin to another. Calculating these paths in real time, has the advantage of being able to modify which smart-bins will be collected. For example, if the time that the collection vehicle can allocate on a specific day is limited, or many smart-bins are full and it is urgent for the waste materials to be collected, the administrator can change the parameter using the Information System; the generated path will only contain bins that are full and the rest will be skipped.

IV. EXPERIMENTAL RESULTS

Kozani, a city in northern Greece, hosted the evaluation phase for the planned smart bin. The tests took place from February 15th to July 15th, 2020. Fig. 4 presents the different places in which we deployed the proposed smart bin prototype. It is difficult to quantify the performance of a project like ours, therefore we cannot provide metrics such as tables and graphs.

We decided to use a single Central Processing Unit (CPU) core in the embedded system we chose for our project. Our tests revealed that there was no bottleneck and that the processing capacity was more than adequate for our project's requirements. Every processor-intensive operation that was required (such as encryption or compression) was completed on time. Residents of Kozani were not hesitant to use the Smart-Bin devices after they were installed, and they were certain that these devices would help with the recycling process. Since Smart-Bins upload measurements and captured images on a regular basis, remote management has proven to be simple and efficient. The Smart-Bins had a very low power consumption because they didn't do any heavy processing. Experiments show that the proposed architecture, which uses stable remote control from the cloud, is a viable solution for both small and large cities.

During evaluation, the project's manager was leading the deployment of the Smart-Bin prototypes and took metrics from the Information System. The Smart-Bins' sensors never captured a temperature or pressure measurement that exceeded the set threshold. We were prepared to remotely activate the Smart-Bins' warning system in the event of an emergency and prevent people from using them.

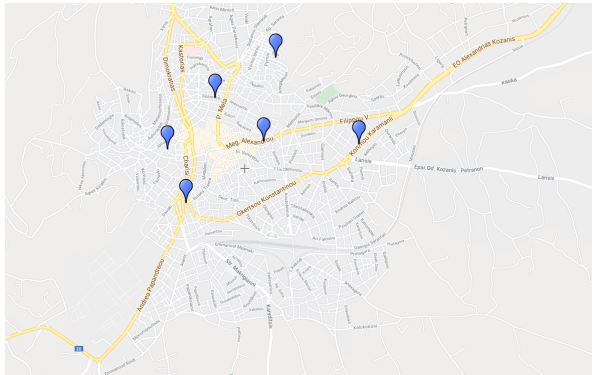


Fig. 4. Blue dots indicate the spots where tests of the smart bin were performed.

V. CONCLUSIONS

Over the last several decades, people have been consuming more, and, as a result, the volume of generated waste has been increasing as well. According to the official statistics, the volume is expected to reach over 2 billion metric tons for the year 2025. Recycling is clearly necessary for a clean and sustainable environment.

We propose a safe cloud-based Smart-Bin for recycling in this paper. People who live in the area have the opportunity to start recycling organic and non-organic materials that would otherwise be thrown away, thanks to the planned IoT device. It measures and photographs items that were disposed inside the bin and alerts future users when the bin contains the maximum number of items. It provides a simple and easy to use notification system that the authorities can utilize to remotely track and control deployed bins by using the cloud. To assess the efficiency of the proposed method, we performed several tests in Western Macedonia, and more specifically in Kozani.

Instead of the Raspberry Pi Zero W, this prototype could be further enhanced in the future by implementing a specifically-designed Printed Circuit Board. More sensors could be used to identify particular harmful substances that are commonly found in the composting process, and the warning system of the smart bins could be automated to prevent or prohibit people from using the Smart-Bins in the event of an emergency. In the future, we are planning to further improve the efficiency of the cloud communication system, compression algorithm and build a data analyzing system. After the project has been finalized, we are also planning to release the source code of the software modules in the public domain; people from around

the world could download the code, make further adjustments and even build their own versions of the smart bin.

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